JACK J. BURLESON

AI / MACHINE LEARNING • DATA SCIENCE

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PROFESSIONAL SUMMARY

Data scientist and ML engineer completing an M.S. in Psychology: Data Science in Human Behavior at UW–Madison (Dec 2026). Builds AI systems end to end: multi-agent LLM document pipelines, calibrated transformer classifiers, LLM evaluation, and behavioral prediction models. Capstone with American Family Insurance on cutting AI inference cost in document processing. Five years of high-volume logistics experience in quality assurance and process improvement.

EDUCATION

University of Wisconsin–Madison *Madison, WI* Expected Dec 2026

M.S. Candidate — Psychology: Data Science in Human Behavior · **GPA 3.85**

- Coursework in machine learning, statistics, predictive modeling, experimental design, and data visualization.

University of Wisconsin–Parkside *Kenosha, WI* Sep 2020 – May 2025

B.S., Psychology · Concentration: Neuroscience · **GPA 3.93**

- Sociology minor; Provost's List (7×), Dean's List (8×), Distinguished Achievement Award (2025), Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Parkside — Undergraduate Researcher, Cognitive Neuroscience Lab *Kenosha, WI*
Jun 2024 – Aug 2025

- Designed and pilot-tested a study of perceived chatbot truthfulness, deception, and user cognitive workload with Dr. Melissa Gregg (code: github.com/Exios66/truth-deception-architecture).
- Operated EEG systems and analyzed neural time-series data for lab research on sleep quality and cognitive load.
- Presented findings at the 2025 Midwestern Psychological Association conference.

Avis Budget Group — Driver *Madison, WI* Jun 2026 – Present

- Move, stage, and safety-check vehicles across a 500+ vehicle rental fleet so cars are ready on time.
- Coordinate vehicle transfers between locations while upholding fleet safety standards.

UW–Madison Recreation & Wellbeing — Building Services *Madison, WI* Jan 2026 – Present

- Keep recreation and wellbeing facilities across campus clean, safe, and ready for use.
- Troubleshoot facility and equipment issues across multiple sites so centers stay safe and open during peak hours.
- Work with operations and event staff to resolve facility issues quickly and minimize disruption.

Meijer Distribution — Logistics Coordinator *Pleasant Prairie, WI* Jul 2020 – Dec 2025

- Promoted from Retail Loader to Logistics Coordinator for consistent quality assurance and process documentation.
- Analyzed quality data for 64 daily outbound trailers across the Midwest network, holding error rates below 1%.
- Partnered with warehouse and transportation teams to resolve quality issues and exceptions before shipment.

TECHNICAL SKILLS

Languages	Python, R, SQL, MATLAB
ML & Statistics	Transformer fine-tuning, probability calibration, cross-validation, SHAP, factor analysis
LLM Engineering	LangGraph, multi-agent pipelines, prompt engineering, LLM evaluation & LLM-as-a-judge
Libraries	pandas, NumPy, SciPy, scikit-learn, XGBoost, PyTorch, Hugging Face Transformers
Tools & Cloud	Git, Docker, pytest, ONNX, FastAPI, Modal, vLLM, Ollama, Parquet, AWS, Azure

PROJECTS

Digital-Mailroom *DSHB Capstone — American Family Insurance (industry partner)*

- Capstone on cutting AI inference cost in document processing: 7 Python packages covering entity extraction, classification, routing, and scalable workflows.
- Designed one LangGraph pipeline that sends each document to the cheapest step that can handle it, joining legal-document routing, prompt experiments, deterministic scoring, dataset management, and reproducible Docker deployment.

Mailroom-ML *ML Training, Evaluation & Fine-Tuning*

- Fine-tuned a ModernBERT document classifier to 95.05% document-type accuracy (307 of 323) and 0.0152 expected calibration error on the held-out test split, with per-head temperature scaling and a confidence gate that decides which documents can skip an LLM.
- Audited the training data, found a label leak (42.8% of fallback filenames named the subclass), removed it, and added a build-time guard; PyTorch training on Modal GPUs, ONNX export for FastAPI serving, and 485 pytest tests in CI.

Mailroom-Corpus-EDA *Data Engineering & Dataset Audit*

- Assembled a 3,302-document, 5-class mailroom corpus from public sources (CUAD, MAUD, SEC EDGAR, Enron email, insurance claims) with stable document IDs, content hashes, per-label provenance, and tracked licenses.
- Built a 7-phase audit pipeline that reruns with one command (integrity, composition, 48 figures, 17 tables), backed by about 200 tests; all 13,753 contract clause annotations match the document text exactly.

RVL-CDIP-Classifier *Computer Vision & Prompt Engineering*

- Built a Python toolkit that converts TIFF/PDF pages to 300-DPI images (with Tesseract OCR) and classifies page images into the 16 RVL-CDIP categories with OpenRouter vision models, at about $0.0004 per image.
- Iterated through 29 registered prompt versions using confusion matrices, misclassification analysis, and cost projections; the final prompt scored 82.6% on a 1,120-image class-balanced evaluation slice.

Psychometric Digital Twin *Behavioral ML Research | UW–Madison*

- Predicted communication-apprehension (PRCA) scores for 241 analyzed survey respondents with Ridge, Elastic Net, k-NN, random forest, histogram gradient boosting, XGBoost, and MLP models, using cross-validation and SHAP feature importance.
- Validated the PRCA's factor structure, then benchmarked LLM "digital twin" personas (vLLM, OpenRouter) against the ML baselines on shared metrics (MAE, band accuracy, F1, AUC): best persona 5.02 vs. 4.49 points of group-score error for the best classical model, with bias analyzed by demographic group.

Langton's Ant *Simulation · TypeScript*

- Built a browser simulator for Langton's ant and other turmites: any rule up to 32 colors and 16 states on an unbounded, chunked grid, with a gallery of 20 published machines and shareable permalinks.
- Kept the engine deterministic (integer-only stepping, seeded randomness) and covered it with 99 passing tests, including a pinned determinism test.

CERTIFICATIONS

CITI Program: Conflicts of Interest 2025

CITI Program: IRB Training for Students and Faculty 2025